

**Michigan State University
Pavement Research Center of Excellence**

**Evaluation of MDOT's Methodologies for both Quantifying
Pavement Distress and Modeling Pavement Performance for Life-
Cycle Cost and Remaining Service Life Estimation Purposes**

Supplementary Document V3.0

User manual for the PDSComp V0.1 Software

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INTRODUCTION

This document is a user's manual for the ***PDSComp*** software written in JAVA. This software loads the MDOT's PMS data (that are in the form of 'csv' files) and processes them to compute the Pavement Distress Score (PDS).

The GUI codes are in the following JAVA files:

- MyLauncher.java
- MainApplication.java
- MainController.java

Compiled version of the GUI is located in the following folder:

- './PDSComp/target/'

In this folder, there is a file "run.bat", which is an executable file and it can be executed by simply double clicking in Windows. For Mac, type the following: "./run.sh"

Once executed, the GUI shown in Figure 1 will appear.

PD/AD (Pre-2019)Data Tab

First tab 'PD/AD (Pre-2019) Data' in the figure is for processing data collected before 2019.

First step is to click "Select PD/AD" button. Once clicked, this will allow you to select multiple csv files. These are the 'Tfiles' shown in **Figure 2**, which includes the PD/AD data collected before 2019.

Next step is to click "Select Pavement List" button shown in **Figure 3**. Once clicked, you'll be asked to select a pavement list that should be in csv format. There are two pavement lists, one for Flexible pavements and one for Rigid pavements. Snapshots of the Flexible and Rigid pavement lists are shown in **Figure 5** and **Figure 6**, respectively. The most important columns are:

- CS
- BMP
- EMP
- DIR
- SURFACE

All the columns above must exist and they should not be blank. Otherwise the program will give error.

Then click the "Compute PDS: Pavement List" button (**Figure 4**). The program will convert all the PD/AD data to SDS equivalent and compute the PDS values for each section in the pavement list. A snapshot of the command prompt while the code is running is shown in **Figure 7**. The outputs of the program is shown in **Figure 8** where separate csv files are generated for each year of PD/AD data and one COMBINED csv file is generated by stacking all the yearly csv files.

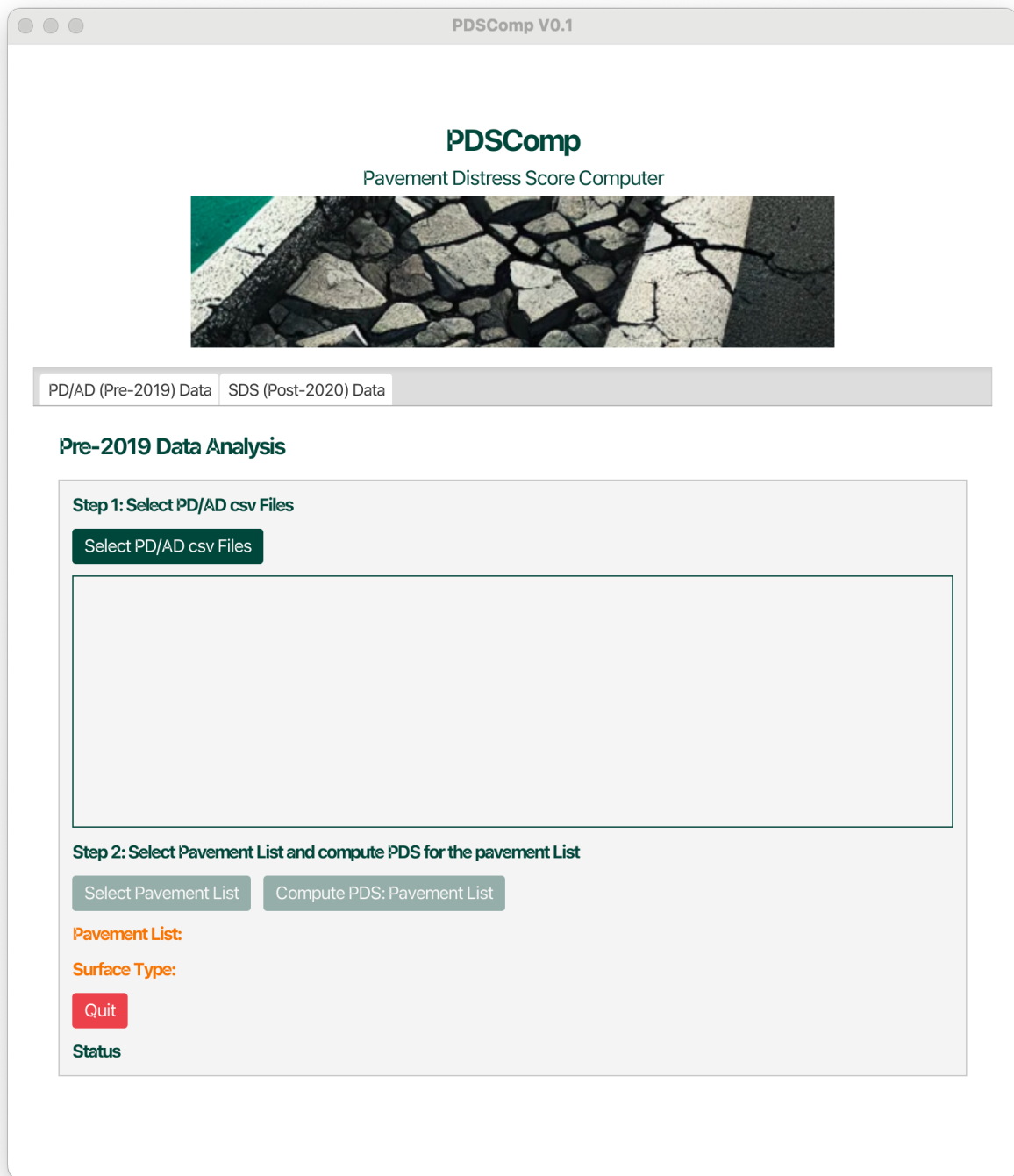


Figure 1 A screenshot of the GUI

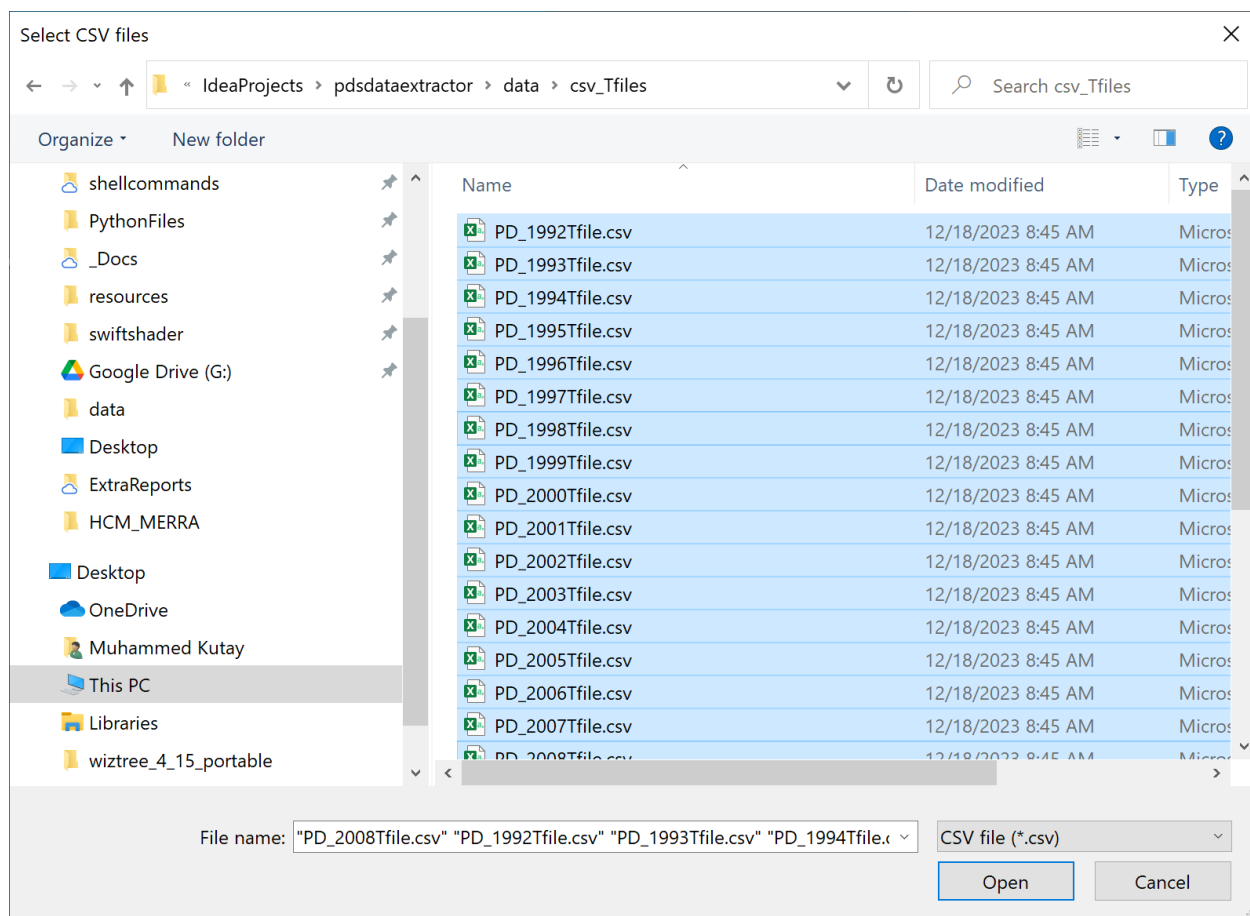


Figure 2 A screenshot of the list of the “Tfiles”, which are the main PMS files

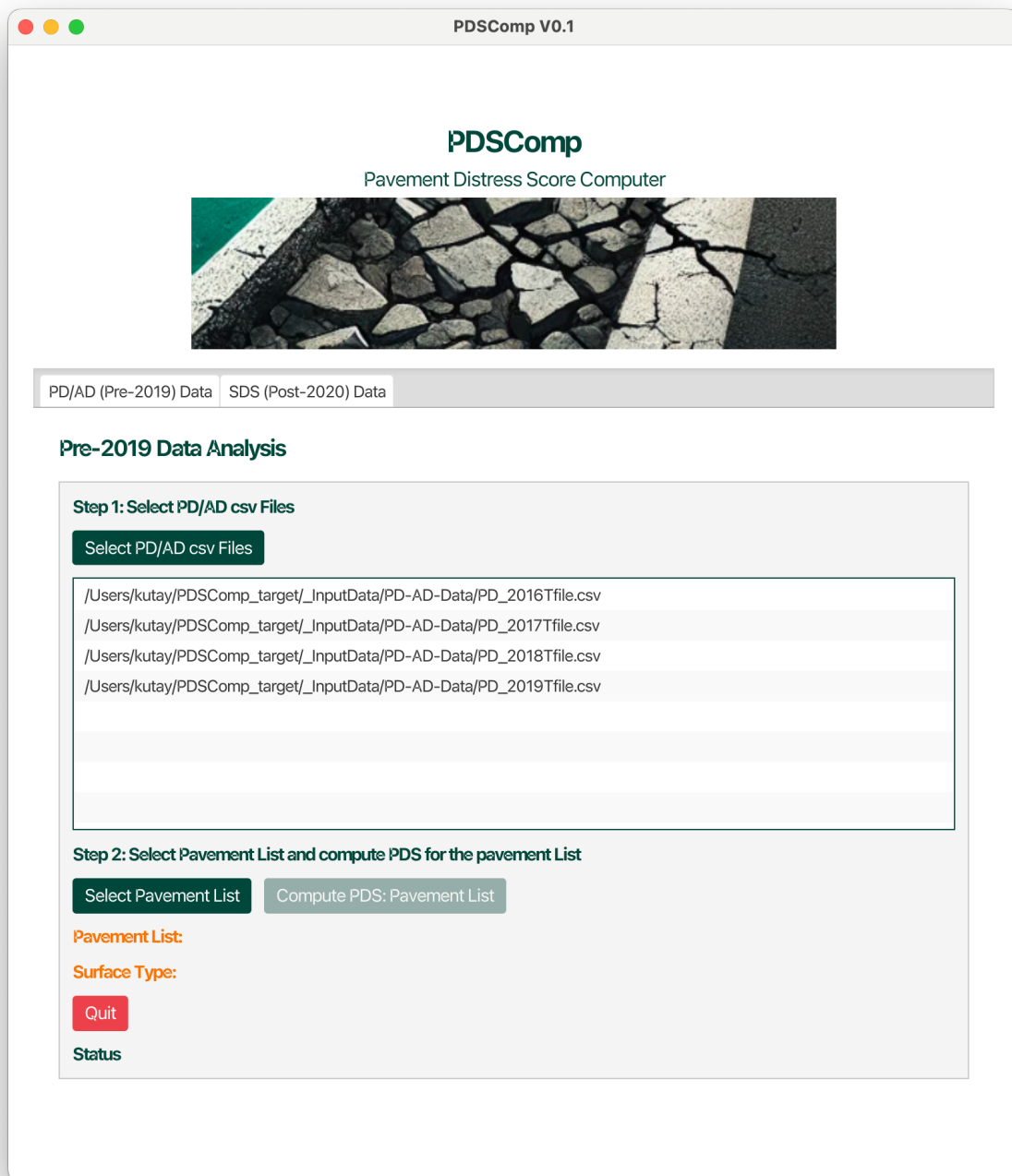


Figure 3 After selection of PD/AD files.

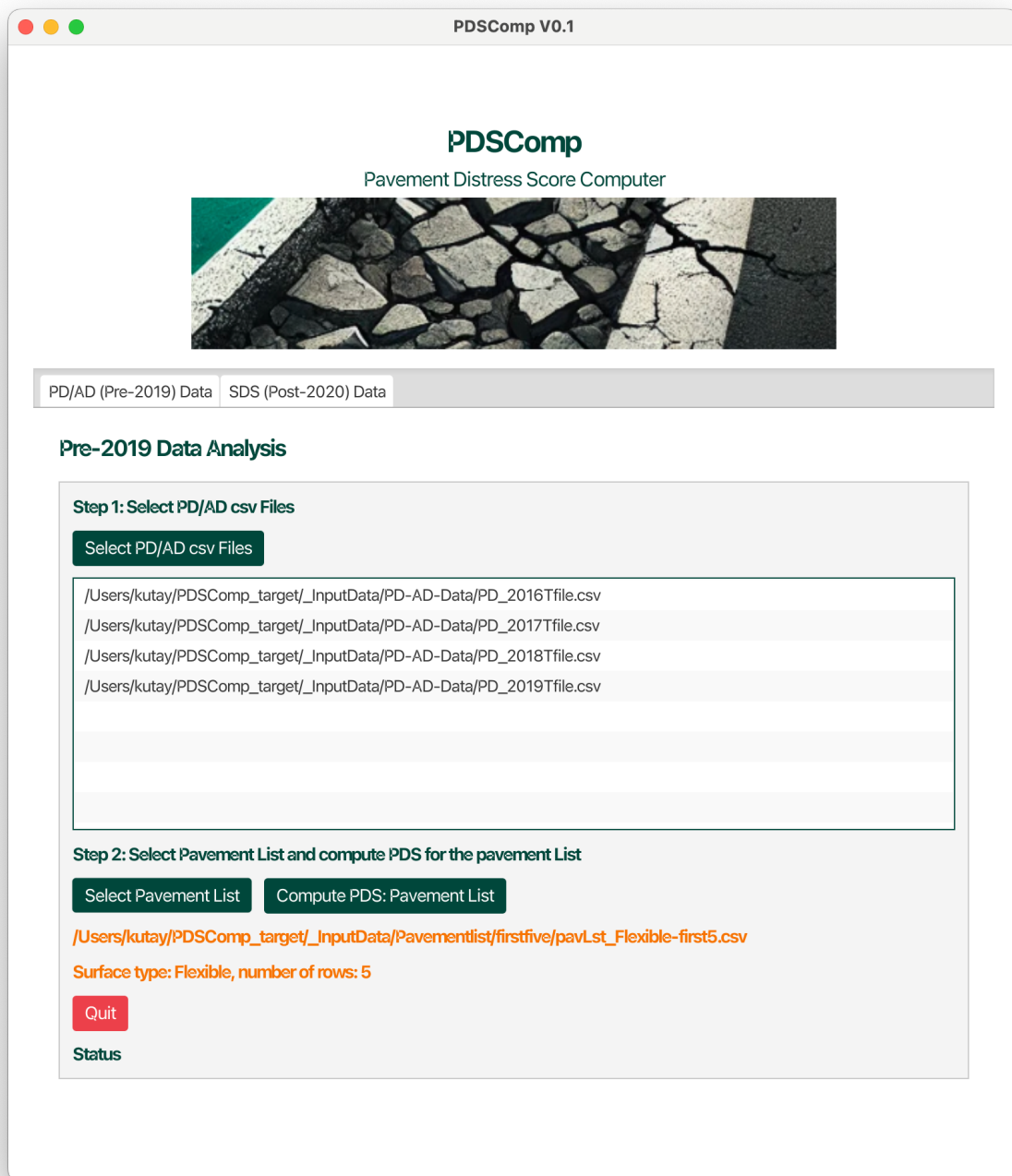


Figure 4 After selection of Pavement List

Finally, clicking “Process Files” will process all the files. A screenshot of the GUI and the command prompt while the program is running is shown in **Figure 7**. The outputs will be saved in the same folder where the input pavement list is located (see **Figure 8**).

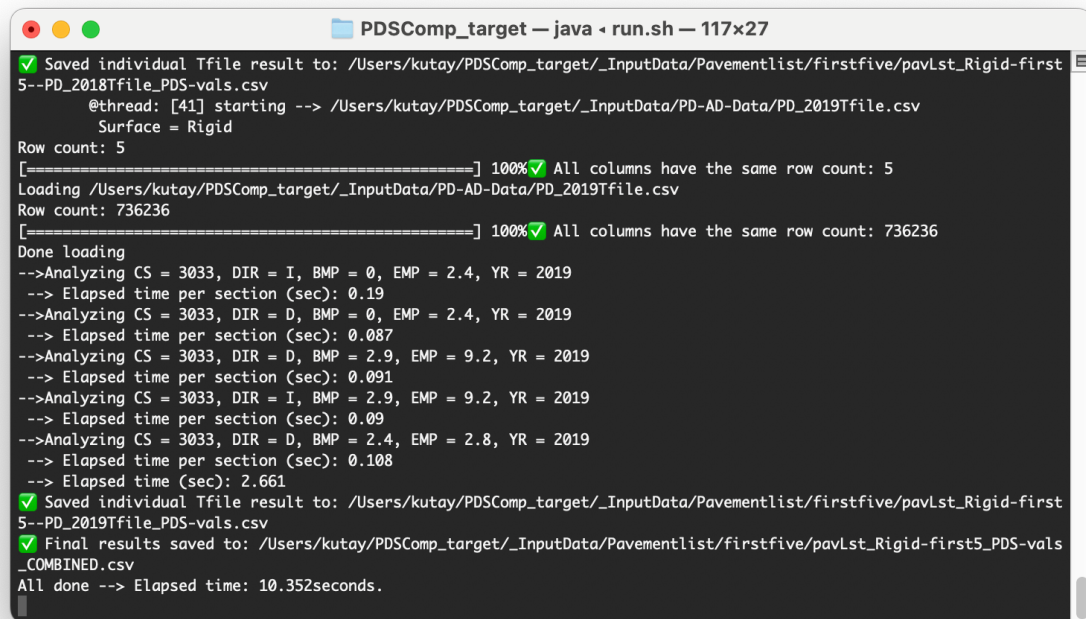
A1 ↕ ✕ ✓ fx | SURFACE

	A	B	C	D	E	F	G	H	
1	SURFACE	PR	PR_BMP	PR_EMP	CS	DIR	BMP	EMP	
2	Flexible	3010827	0	3.904	1011	I	0	3.9	
3	Flexible	3010827	4.218	10.953	1011	I	4.3	10.953	
4	Flexible	1725010	0	4.372	1012	I	0	4.3	
5	Flexible	1725010	0.115	4.349	1012	I	0.2	4.3	
6	Flexible	3010070	0	0.229	1012	I	9.852	10.081	
7									
8									
9									
10									
11									

Figure 5 A screenshot of the Flexible pavement list

	A	B	C	D	E	F	G	H	
1	SURFACE	PR	PR_BMP	PR_EMP	CS	DIR	BMP	EMP	
2	Rigid	784007	0	2.405	3033	I	0	2.4	
3	Rigid	784008	0	2.405	3033	D	0	2.4	
4	Rigid	784008	2.833	9.24	3033	D	2.9	9.2	
5	Rigid	784007	2.864	9.241	3033	I	2.9	9.2	
6	Rigid	784008	2.398	2.833	3033	D	2.4	2.8	
7									
8									

Figure 6 A screenshot of the Rigid pavement list



```
PDSComp_target - java - run.sh - 117x27
✓ Saved individual Tfile result to: /Users/kutay/PDSComp_target/_InputData/Pavementlist/firstfive/pavLst_Rigid-first
5--PD_2018Tfile_PDS-vals.csv
@thread: [41] starting --> /Users/kutay/PDSComp_target/_InputData/PD-AD-Data/PD_2019Tfile.csv
Surface = Rigid
Row count: 5
[=====] 100% ✓ All columns have the same row count: 5
Loading /Users/kutay/PDSComp_target/_InputData/PD-AD-Data/PD_2019Tfile.csv
Row count: 736236
[=====] 100% ✓ All columns have the same row count: 736236
Done loading
-->Analyzing CS = 3033, DIR = I, BMP = 0, EMP = 2.4, YR = 2019
--> Elapsed time per section (sec): 0.19
-->Analyzing CS = 3033, DIR = D, BMP = 0, EMP = 2.4, YR = 2019
--> Elapsed time per section (sec): 0.087
-->Analyzing CS = 3033, DIR = D, BMP = 2.9, EMP = 9.2, YR = 2019
--> Elapsed time per section (sec): 0.091
-->Analyzing CS = 3033, DIR = I, BMP = 2.9, EMP = 9.2, YR = 2019
--> Elapsed time per section (sec): 0.09
-->Analyzing CS = 3033, DIR = D, BMP = 2.4, EMP = 2.8, YR = 2019
--> Elapsed time per section (sec): 0.108
--> Elapsed time (sec): 2.661
✓ Saved individual Tfile result to: /Users/kutay/PDSComp_target/_InputData/Pavementlist/firstfive/pavLst_Rigid-first
5--PD_2019Tfile_PDS-vals.csv
✓ Final results saved to: /Users/kutay/PDSComp_target/_InputData/Pavementlist/firstfive/pavLst_Rigid-first5_PDS-vals
_COMBINED.csv
All done --> Elapsed time: 10.352seconds.
```

Figure 7 A screenshot of the command prompt while the program is running.

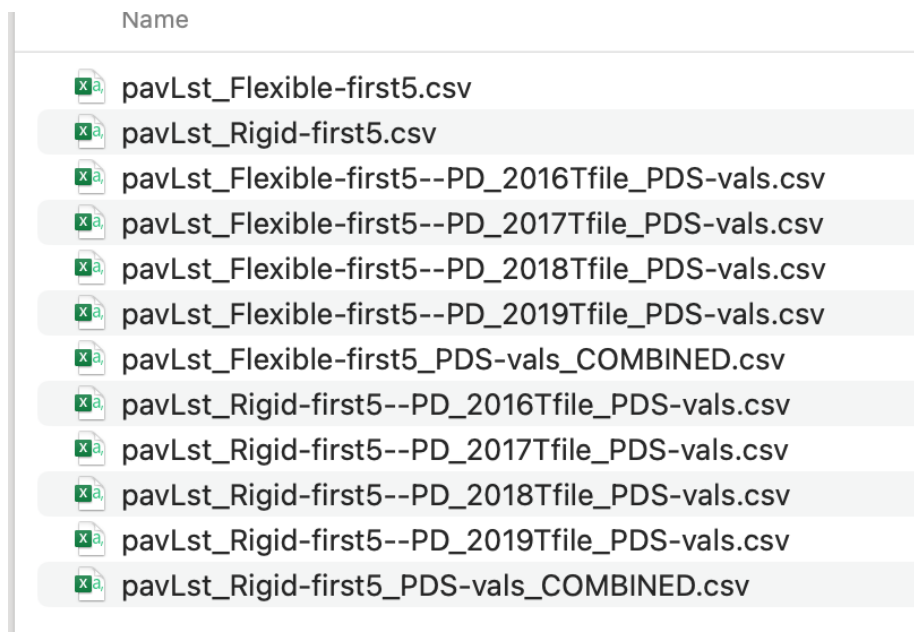


Figure 8 A screenshot of the outputs.

SDS (Post-2020) Data Tab

When the user clicks the SDS (Post-2020) Data tab, the GUI in Figure 9 becomes visible. Here, First user clicks the 'Select SDS csv files' button. Once user selects the SDS data, next step is computation of PDS values for every row of the SDS data in the list. A screenshot of the SDS (Post-2020) Data tab after clicking the Select SDS csv files button is shown in Figure 10. At this point, user clicks the 'Compute PDS: all rows' button to compute the PDS values for every 0.1 mile or 0.01 mile section that is in the SDS csv files selected.

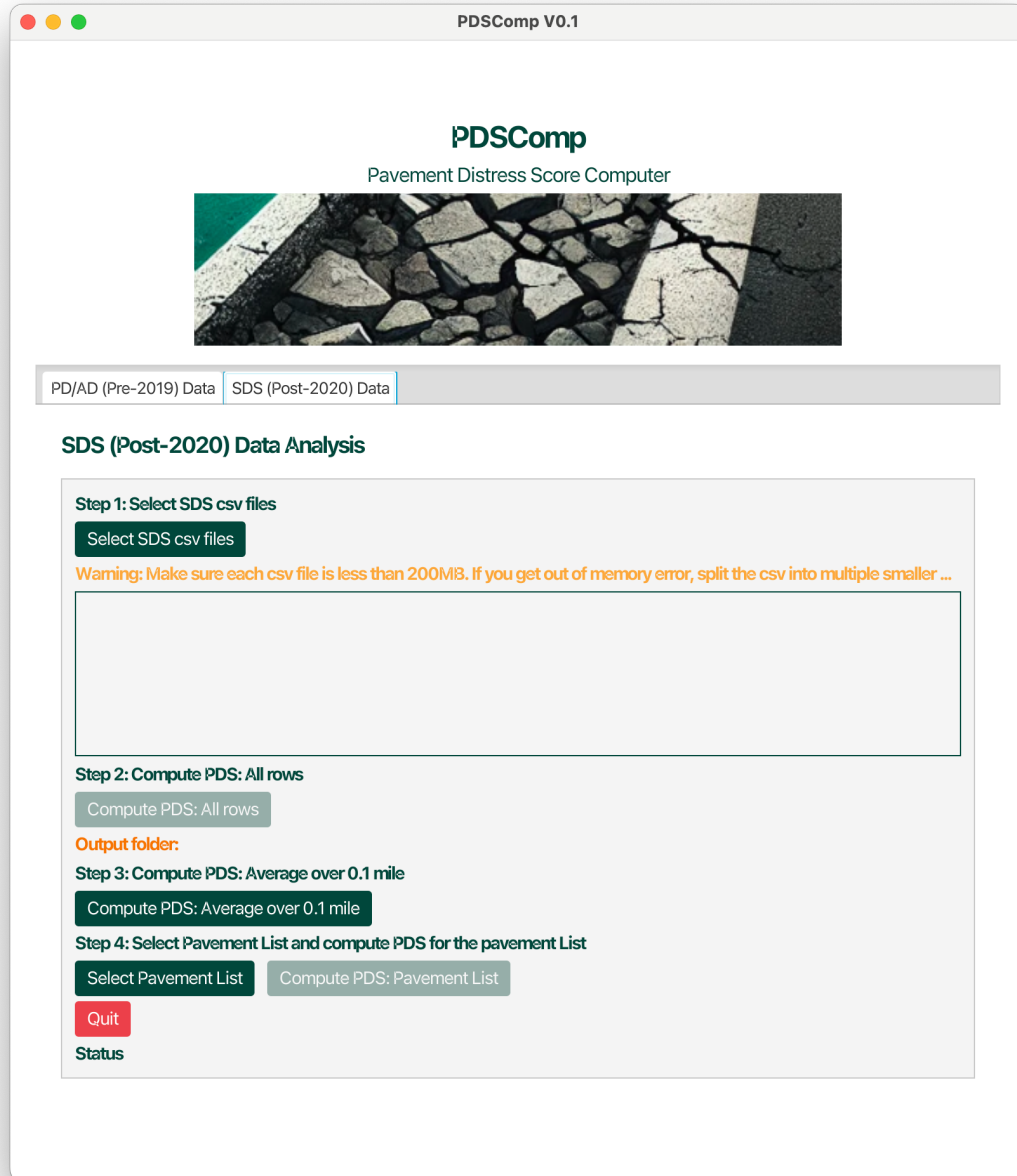


Figure 9 A screenshot of the SDS (Post-2020) Data Tab.



Figure 10 A screenshot of the SDS (Post-2020) Data tab after clicking the Select SDS csv files button.

Next step is to compute average 0.1 mile segment PDS data from the 0.01 mile data. This is done by clicking the 'Compute PDS:Average over 0.1 mile' button. This step is not necessary for the years where the original SDS data is already in 0.1 mile increments.

Finally next step is to compute the PDS values for the selected pavement sections. User clicks the 'Select Pavement List' button and selects the files shown in Figure 5 and Figure 6. The most important columns are:

- PR
- PR_BMP
- PR_EMP
- SURFACE

All other columns can be empty because at this point only these are used.

Finally, user clicks Copute PDS: Pavement List button to compute the average PDS values over the given BMP/EMP ranges in the pavement list.